Prioritised Revision Guide

Cambridge Part II Soft Condensed Matter

Based on analysis of all 12 past papers (2014–2025)
and the course handout (Souslov, 22 lectures)

How to use this guide

Topics are ranked by exam frequency, weighted towards the most recent papers. The **green percentages** show appearance rates across 12 papers (no paper in 2020 due to COVID). Tier 1 topics are near-certain; prepare these first.

TIER 1 — Near-Certain to Appear (65%+ frequency)

1. Numerical / Order-of-Magnitude Estimates 92%

At least one short question every year requires a quick numerical calculation. Practise these until they are automatic.

Standard estimates to know:

- Reynolds number: $\mathrm{Re} = \rho v L/\eta$. Bacterium ($\sim 2\,\mu$m, $25\,\mu$m/s): $\mathrm{Re} \sim 5 \times 10^{-5}$. Whale: $\mathrm{Re} \sim 3 \times 10^{8}$.
- Diffusion time: $t \sim x^2/D$ with $D = k_\mathrm{B}T/(6\pi\eta a)$. Protein ($a = 3\,$nm) to diffuse $1\,\mu$m in water: $t \sim 0.5\,$ms.
- Stokes terminal velocity: $v = 2(\rho_p - \rho_f)ga^2/(9\eta)$. Know how to do this for iron oxide particles, haemoglobin in ultracentrifuge.
- Debye length: $\kappa^{-1} = \sqrt{\varepsilon_0\varepsilon_r k_\mathrm{B}T/(2z^2e^2n_\infty)}$. In 0.01 M NaCl: $\kappa^{-1} \approx 3\,$nm. Scale with valency and concentration.
- Capillary length: $\ell_c = \sqrt{\gamma/(\rho g)} \approx 2.7\,$mm for water–air.
- Crosslink density from rubber modulus: $n_c = G/(k_\mathrm{B}T)$.

Tested: Every year except 2020

2. Polymer Chain Statistics 75%

Ideal chain (random walk)

$\langle R^2\rangle = Nb^2$, where N = number of Kuhn monomers, b = Kuhn length. End-to-end distribution:

$$P(\vec{R}) = \left(\frac{3}{2\pi N b^2}\right)^{3/2} \exp\left(-\frac{3R^2}{2Nb^2}\right)$$

Entropy: $S(R) = \mathrm{const} - 3k_\mathrm{B}R^2/(2Nb^2)$. Entropic spring constant: $k = 3k_\mathrm{B}T/(Nb^2)$.

Tested: 2014, 2015, 2017, 2018, 2019, 2022, 2023, 2024, 2025

Self-avoiding walk

$R \sim N^\nu b$ with Flory exponent $\nu = 3/5$ (3D good solvent), $\nu = 1/2$ (θ-solvent), $\nu = 3/4$ (2D). **Derive** $\nu = 3/5$ by minimising $F = k_B T R^2/(Nb^2) + k_B T v N^2/(R^3)$.

Tested: 2017, 2018, 2023

Worm-like chain

$\langle R^2 \rangle = 2\ell_p L \left[1 - \frac{\ell_p}{L}\left(1 - e^{-L/\ell_p}\right)\right]$. Limits: $L \gg \ell_p$: $\langle R^2 \rangle = 2\ell_p L = N_K \ell_K^2$ (random walk with $\ell_K = 2\ell_p$). $L \ll \ell_p$: $\langle R^2 \rangle = L^2$ (rigid rod). Persistence length of dsDNA: $\ell_p \approx 50\,\text{nm}$.

Tested: 2014, 2018, 2023

Overlap concentration

$\phi^* \sim N^{1-3\nu}$. Good solvent ($\nu = 3/5$): $\phi^* \sim N^{-4/5}$. θ-solvent ($\nu = 1/2$): $\phi^* \sim N^{-1/2}$.

Tested: 2017, 2018, 2019, 2024, 2025

3. Viscoelasticity & Complex Modulus 67%

Complex modulus — tested almost every other year

$G^*(\omega) = G'(\omega) + iG''(\omega)$. G' = storage (elastic); G'' = loss (viscous). Hookean solid: $G^* = G_0$. Newtonian fluid: $G^* = i\omega\eta$.

$$G^*(\omega) = i\omega \int_0^\infty G(t)\, e^{-i\omega t} dt$$

Maxwell model (spring + dashpot in series): $G(t) = G_0\, e^{-t/\tau}$, $\tau = \eta/G_0$.

$$G^*(\omega) = \frac{i\omega G_0 \eta}{G_0 + i\omega\eta} = \frac{G_0 \omega^2\tau^2}{1+\omega^2\tau^2} + i\frac{G_0\omega\tau}{1+\omega^2\tau^2}$$

Low ω: $G' \propto \omega^2$, $G'' \propto \omega$. High ω: $G' \to G_0$, $G'' \to 0$. Crossover at $\omega\tau = 1$.

Kelvin–Voigt model (spring + dashpot in parallel): $G^*(\omega) = G_0 + i\omega\eta_0$. No stress relaxation. Used for creep.

Zener model (Maxwell in parallel with spring G_1): $G^*(\omega) = G_1(1 + i\omega\tau_\varepsilon)/(1 + i\omega\tau_\sigma)$, with $\tau_\sigma = \eta/G_2$, $\tau_\varepsilon = \eta(G_1 + G_2)/(G_1 G_2)$.

Tested: 2014, 2015, 2017, 2018, 2021, 2022, 2024, 2025

4. Debye–Hückel Screening & Electrostatics 67%

$$\kappa^2 = \frac{2z^2e^2 n_\infty}{\varepsilon_0\varepsilon_r k_B T}, \qquad \kappa^{-1} \propto \frac{1}{z\sqrt{n_\infty}}$$

Linearised Poisson–Boltzmann: $\nabla^2\phi = \kappa^2\phi$. For a sphere of radius a and charge Q: $\phi(r) = \frac{Q}{4\pi\varepsilon_0\varepsilon_r}\frac{e^{-\kappa(r-a)}}{r(1+\kappa a)}$.

Key scalings: κ^{-1} halves when concentration quadruples (monovalent). Replacing NaCl with $AlCl_3$: z changes from 1 to 3, κ increases by factor $\sqrt{6}$.

Tested: 2015, 2016, 2017, 2018, 2019, 2022, 2023, 2025

5. Flory–Huggins Theory & Polymer Mixing 50%

Free energy per lattice site:

$$f = k_B T \left[\frac{\phi}{N}\ln\phi + (1-\phi)\ln(1-\phi) + \chi\phi(1-\phi)\right]$$

First term: polymer entropy (suppressed by $1/N$). Second: solvent entropy. Third: interaction energy.

Spinodal: $\partial^2 f/\partial\phi^2 = 0 \Rightarrow \chi(\phi) = \frac{1}{2}\left(\frac{1}{\phi N} + \frac{1}{1-\phi}\right)$.

Critical point: $\phi_c = 1/(1+\sqrt{N}) \approx N^{-1/2}$, $\chi_c = \frac{1}{2}\left(1 + 1/\sqrt{N}\right)^2 \approx 1/2 + N^{-1/2}$.

Osmotic pressure: $\Pi = -k_\mathrm{B}T\left[\ln(1-\phi) + (1-1/N)\phi + \chi\phi^2\right]/v_0$. For $\phi \to 0$: $\Pi \approx k_\mathrm{B}T\left[\phi/N + (1/2-\chi)\phi^2 +\right.$ At $\chi = 1/2$: $\Pi \propto \phi$ (ideal). $\chi < 1/2$: good solvent (repulsive second virial). $\chi > 1/2$: poor solvent (phase separation).

Sketch the phase diagram: T vs ϕ, marking spinodal, binodal, critical point, and regions of stability/metastability/instability. Know spinodal decomposition vs nucleation and growth.

Tested: 2014, 2016, 2017, 2019, 2021, 2025

6. DLVO Theory & Colloidal Stability 50%

Total interaction: $V(r) = V_\mathrm{vdW}(r) + V_\mathrm{elec}(r)$. Van der Waals: attractive, $V_\mathrm{vdW} \propto -1/r^6$ (atom–atom) or $-A/(12h)$ per unit area (flat surfaces). Electrostatic: repulsive, $V_\mathrm{elec} \propto \mathrm{e}^{-\kappa r}$.

Sketch the combined potential: deep primary minimum at contact, energy barrier at intermediate r, shallow secondary minimum at large r. As salt increases → barrier decreases → coagulation.

Stabilisation mechanisms: charge (electrostatic double-layer), steric (grafted polymers), depletion (small polymer additives create attractive osmotic force).

Tested: 2014, 2018, 2019, 2021, 2022, 2024

TIER 2 — High Probability (35–55% frequency)

7. Reynolds Number & Low-Re Physics 50%

$\mathrm{Re} = \rho v L/\eta$. Stokes equation ($\mathrm{Re} \ll 1$): $0 = -\nabla p + \eta\nabla^2\vec{v}$. Properties: instantaneous, linear, time-reversible. Scallop theorem: reciprocal motion cannot produce net translation at low Re. Bacteria use rotating flagella or non-reciprocal cilia.

Tested: 2015, 2016, 2017, 2018, 2019, 2023

8. Surface Tension, Wetting & Young's Equation 42%

$\gamma_{SV} = \gamma_{SL} + \gamma_{LV}\cos\theta$. Contact angle θ: hydrophobic ($\theta > 90$), hydrophilic ($\theta < 90$). **Capillary rise:** $h = 2\gamma\cos\theta/(\rho g R)$.

Tested: 2014, 2016, 2017, 2021, 2022

9. Amphiphiles & Self-Assembly 42%

Packing parameter: $p = v/(a_0 l_c)$. $p < 1/3$: spherical micelles. $1/3 < p < 1/2$: cylindrical. $1/2 < p < 1$: bilayers/vesicles.

CMC derivation: Chemical potential $\mu = \mu_\alpha^0/\alpha + (k_\mathrm{B}T/\alpha)\ln(x_\alpha/\alpha)$. Equating monomer and aggregate chemical potentials gives $x_\alpha = \alpha[x_1 \mathrm{e}^{(\mu_1^0-\mu_\alpha^0)/k_\mathrm{B}T}]^\alpha$. CMC $\approx \mathrm{e}^{-(\mu_1^0-\mu_\alpha^0)/k_\mathrm{B}T}$.

Tested: 2015, 2016, 2019, 2022, 2023

10. Membrane Fluctuations & Bending Rigidity 33%

$\langle |h_q|^2 \rangle = \frac{k_B T}{A(\sigma q^2 + \kappa q^4)}$. σ = tension, κ = bending rigidity. Small length scales: bending dominates (κq^4). Crossover at $q^* = \sqrt{\sigma/\kappa}$.

Tested: 2014, 2017, 2023, 2024

11. Stokes Drag & Sedimentation 42%

$F = 6\pi\eta a v$. Stokes–Einstein: $D = k_B T/(6\pi\eta a)$. Terminal velocity: balance $6\pi\eta a v = \frac{4}{3}\pi a^3 \Delta\rho\, g$. Sedimentation equilibrium: $\langle h \rangle = k_B T/(m_b g)$.

Tested: 2014, 2016, 2018, 2019

12. Poiseuille Flow 25%, but tested 2018, 2021, 2023

Circular pipe: $Q = \pi a^4 \Delta p/(8\eta L)$. Flow rate $\propto a^4$ (doubling radius $\to$ 16$\times$ flow). Parallel plates: $Q = w h^3 \Delta p/(12\eta L)$.

Tested: 2018, 2021, 2023

TIER 3 — Moderate Probability (15–30% frequency)

13. Blob Model & Scaling 17%

Chain in a tube/on a surface: subdivide into blobs of size ξ, each containing g monomers with $\xi \sim g^\nu b$. Free energy per blob $\sim k_B T$. Brush height in good solvent: $h \sim N b(\sigma b^2)^{1/3}$.

Tested: 2016, 2022

14. Electrophoresis 17%

$v = ZeE/(6\pi\eta R)$ for a charged sphere. Gel electrophoresis separates DNA by size (Ogston sieving + reptation). Bands: distance $\propto \log(1/N)$ at constant time.

Tested: 2019, 2021

15. Coil–Helix Transition 17%, rising

Helix fraction: $f_h = \frac{1}{2} + \frac{s-1}{2\sqrt{(s-1)^2 + 4s\sigma}}$, with $s = e^{-\beta\epsilon}$, σ = cooperativity. Small σ: sharp transition (high cooperativity). Large σ: gradual.

Tested: 2018, 2025

16. Charged Polymers / Polyelectrolytes 8%, but 2024

Coulomb stretching of a chain with end charges $+Ze$: minimise $F = 3k_B T R^2/(2Nb^2) + Z^2 e^2/(4\pi\varepsilon R)$. Gives $R \propto (Z^2 l_B N b^2)^{1/3}$. Debye–Hückel screening reduces extension.

Tested: 2024

17. Janus Particles / Pickering Emulsions 8%, but 2025

Adsorption energy of a spherical particle (radius a) at a fluid interface: $\Delta E \sim \pi a^2 \gamma$. Becomes $\sim k_B T$ when $a \sim \sqrt{k_B T/(\pi\gamma)} \sim 1\,\mathrm{nm}$. Much larger than molecular surfactants $\to$ irreversible adsorption.

Tested: 2025

Cross-Cutting Skills (tested every year)

Five skills that earn marks across all questions

1. **Dimensional analysis & estimates.** Most short questions are really asking you to plug into a formula and get a number. Know $k_B T \approx 4 \times 10^{-21}$ J at 300 K, $\eta_{\text{water}} = 10^{-3}$ Pa s, $\gamma_{\text{water}} = 72$ mN/m, $e = 1.6 \times 10^{-19}$ C.
2. **Sketching.** Be prepared to sketch: DLVO potential vs r, Flory–Huggins phase diagram (T vs ϕ), $G'(\omega)$ and $G''(\omega)$ for Maxwell/Zener models, capillary number/contact angle diagrams, micelle aggregation number distributions.
3. **Deriving from free energy.** Many long questions give you $f(\phi)$ or $F(R)$ and ask you to minimise, differentiate, or integrate. Master: $\partial f/\partial\phi = 0$ for equilibrium, $\partial^2 f/\partial\phi^2 = 0$ for spinodal, $\Pi = -\partial F/\partial V$ for osmotic pressure.
4. **Scaling arguments.** The examiners love de Gennes-style scaling: $R \sim N^\nu$, $\phi^* \sim N^{1-3\nu}$, brush height $h \sim N\sigma^{1/3}$, etc. Know how to construct these from blob arguments.
5. **Connecting models to experiments.** Questions often ask "what experimental technique would you use?" or "sketch what you would measure." Know: rheometry (G* from oscillatory shear), DLS (diffusion coefficient), osmometry (Π vs c gives M_w), capillary viscometry, electrophoresis, AFM force curves.